

CareerMatch AI – Data Engineer Role Guide

Synthetic/Generalized Career Knowledge – Hackathon Prototype

1. Role Overview

A Data Engineer builds and maintains the systems that move and prepare data for analysts, data scientists, and applications. The focus is on reliability, scale, and correctness of data pipelines rather than modeling. This is generalized educational content for a hackathon prototype.

2. Responsibilities

- Design and build data pipelines that move data between systems
- Model and organize data for efficient storage and querying
- Ensure data quality, consistency, and reliability
- Work with databases and, where relevant, distributed data systems
- Support analysts and data scientists with clean, accessible data

3. Required Skills

Python

What it is: A general-purpose, beginner-friendly programming language widely used for data work, automation, and AI development.

Why it matters: It is the primary language for machine learning, data analysis, and AI tooling because of its simple syntax and large ecosystem of libraries.

How it's used in this role: Used to load data, build models, write automation scripts, and connect components of an AI pipeline.

Beginner expectation: Can write scripts using variables, loops, conditionals, functions, and lists/dictionaries.

SQL

What it is: A query language used to read, filter, and aggregate data stored in relational databases.

Why it matters: Most real-world data lives in databases, so extracting and shaping it correctly is a foundational skill for any data-related role.

How it's used in this role: Used to pull structured datasets from a company database before analysis or model training.

Beginner expectation: Can write SELECT, WHERE, GROUP BY, and simple JOIN queries.

Databases

What it is: Organized systems for storing, managing, and retrieving data.

Why it matters: Almost every application and analysis depends on data stored in a database.

How it's used in this role: Used as the foundation for storing data used in analysis or applications.

Beginner expectation: Understands what a database is and how tables relate to each other.

Data Modeling

What it is: The process of designing how data is structured and related within a database or system.

Why it matters: Good data models make data easier to query, maintain, and scale.

How it's used in this role: Used to design the structure of a data warehouse or application database.

Beginner expectation: Understands basic table relationships and normalization concepts.

ETL

What it is: Extract, Transform, Load: the process of moving data from source systems into a usable format for analysis.

Why it matters: Reliable data pipelines are the foundation that all downstream analysis and machine learning depend on.

How it's used in this role: Used to move and prepare data before it reaches analysts or data scientists.

Beginner expectation: Understands the ETL process conceptually and can write a simple data pipeline script.

Data Pipelines

What it is: Automated sequences of steps that move and transform data from source to destination.

Why it matters: Pipelines make data processing repeatable, reliable, and scalable.

How it's used in this role: Used to feed clean, reliable data into analytics or machine learning systems.

Beginner expectation: Can write a simple script that moves and transforms data on a schedule.

4. Database Technologies

PostgreSQL

What it is: An open-source relational database management system.

Why it matters: It is one of the most widely used production-grade relational databases.

How it's used in this role: Used to store and query structured application or analytical data.

Beginner expectation: Can create tables and run basic queries.

MySQL

What it is: A popular open-source relational database management system.

Why it matters: It is widely used in web applications and is a common choice for learning relational databases.

How it's used in this role: Used to store structured data for applications and analytics.

Beginner expectation: Can create tables and run basic queries.

NoSQL concepts

What it is: Non-relational database models such as document, key-value, or graph stores.

Why it matters: Some data, especially unstructured or rapidly changing data, fits non-relational models better.

How it's used in this role: Used when data does not fit neatly into rigid table structures.

Beginner expectation: Understands the difference between relational and non-relational databases.

5. Big Data Concepts

Big Data

What it is: Datasets and systems that are too large or fast-moving for traditional single-machine processing.

Why it matters: Many organizations need distributed systems to store and process data at scale.

How it's used in this role: Used conceptually when designing systems that handle large-scale data.

Beginner expectation: Understands why distributed processing is needed for very large datasets.

Spark concepts

What it is: Concepts behind Apache Spark, a distributed computing framework for processing large datasets.

Why it matters: It allows data processing to scale across multiple machines when data is too large for one computer.

How it's used in this role: Used conceptually when designing systems for very large-scale data processing.

Beginner expectation: Understands why distributed processing is needed and the basic Spark execution model.

6. Cloud Concepts

Cloud Computing

What it is: The delivery of computing resources such as storage and servers over the internet.

Why it matters: Most production AI and data systems run on cloud infrastructure rather than local machines.

How it's used in this role: Used to host datasets, models, or applications at scale.

Beginner expectation: Understands what cloud services are used for at a conceptual level.

7. Docker

Docker

What it is: A platform for packaging applications and their dependencies into portable containers.

Why it matters: It ensures an application runs consistently across different machines and environments.

How it's used in this role: Used to package a model or application so it can be deployed reliably.

Beginner expectation: Can write a simple Dockerfile and run a container.

8. APIs

APIs

What it is: Interfaces that allow different software systems to communicate with each other.

Why it matters: APIs connect models, data sources, and applications into a working system.

How it's used in this role: Used to connect a trained model to a frontend or another application.

Beginner expectation: Understands what an API is and can call a simple public API.

9. Education

Commonly pursued by graduates in Computer Science, Information Technology, or related engineering fields, often with strong database and systems fundamentals.

10. Entry-Level Expectations

- Can write efficient SQL queries and design a simple relational schema
- Understands the basic ETL process and can build a simple pipeline script
- Aware of why distributed systems are needed for very large datasets
- Comfortable with basic version control and reproducible project setup
- These expectations describe a general, entry-level readiness bar used only for hackathon prototyping. They are illustrative and not tied to any real company's actual hiring bar, vacancy, or deadline.

11. Skill Priority

High Priority	Medium Priority	Low Priority
• SQL • Python • ETL • Data Modeling	• Databases • Data Pipelines • Cloud Computing	• Big Data • Spark concepts • Docker

12. Common Skill Gaps

- Comfortable with basic SQL but limited schema design experience
- Little exposure to automated or scheduled data pipelines
- Unfamiliarity with distributed processing concepts
- Limited hands-on cloud platform experience

13. Portfolio Projects

1. Automated Weather Data Pipeline

Problem: Build a pipeline that regularly ingests, cleans, and stores weather data.

Skills used: Python, SQL, ETL

Difficulty: Intermediate

2. E-commerce Data Warehouse Design

Problem: Design a schema to store and query orders, products, and customers efficiently.

Skills used: Data Modeling, SQL, PostgreSQL

Difficulty: Intermediate

14. Interview Topics

- Relational database design and normalization
- ETL pipeline design and scheduling
- SQL query optimization
- Basic distributed systems concepts

15. Interview Questions

- 1 What is the difference between ETL and ELT?
- 2 How would you design a schema for an e-commerce platform?
- 3 What steps would you take if a daily data pipeline failed silently?
- 4 When would you choose a NoSQL database over a relational one?

16. Preparation Roadmap

- 1 SQL and relational database fundamentals
- 2 Python for scripting and automation
- 3 Data modeling
- 4 Building ETL pipelines
- 5 Cloud basics
- 6 Big data concepts (optional, advanced)

17. RAG Keywords

data engineer, ETL pipeline, data modeling, SQL for data engineering, entry-level data engineer, data pipelines

Document Metadata

Job ID	DE001
Role	Data Engineer
Category	Data Engineering
Experience Level	Entry Level
Data Status	Synthetic/Generalized